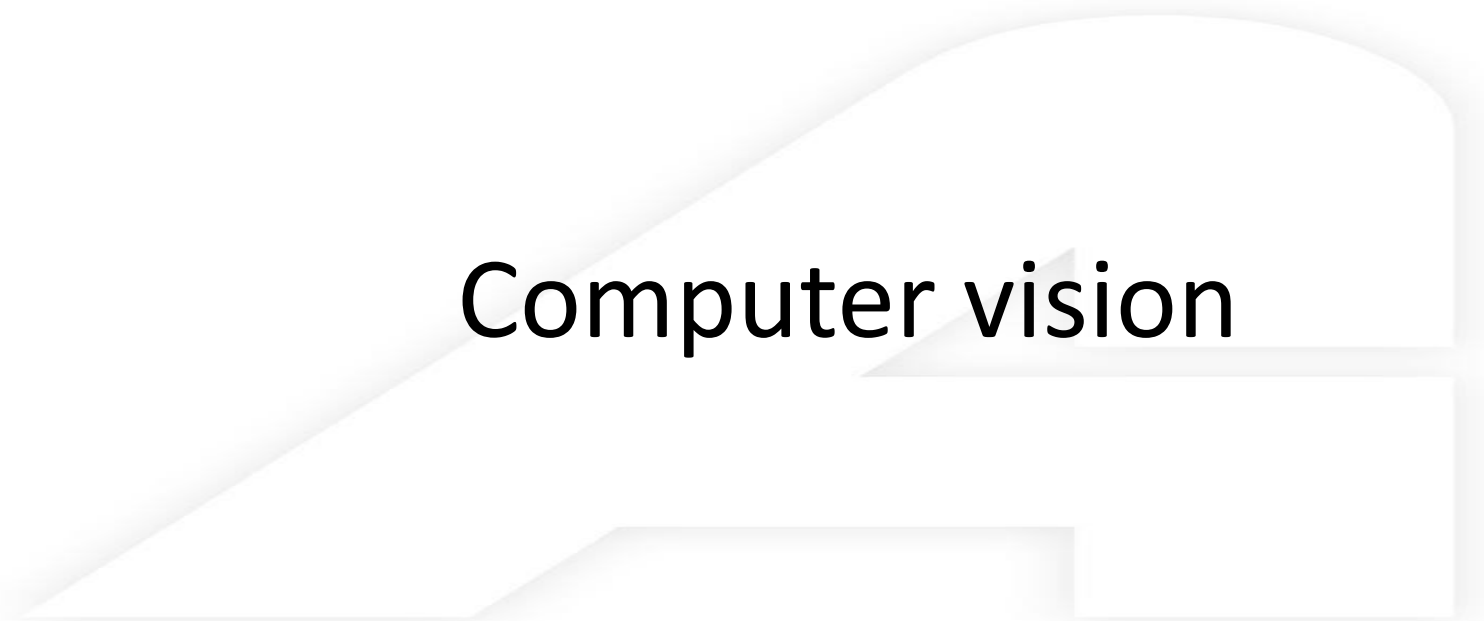




Computer vision

Agenda:

- 1- Definition of Image processing.
- 2- Definition of computer vision.
- 3- Image and it's type .
- 4- color-spaces models.
- 5- Image Digitization.
- 6- Coordinate Transformation.
- 7- Histogram & Histogram Equalization.
- 9- Noise & Filters.

What is the image processing ?

Image processing is a method to perform some operations on an image, in order to get an enhanced image or to extract some useful information from it

Example of operations :

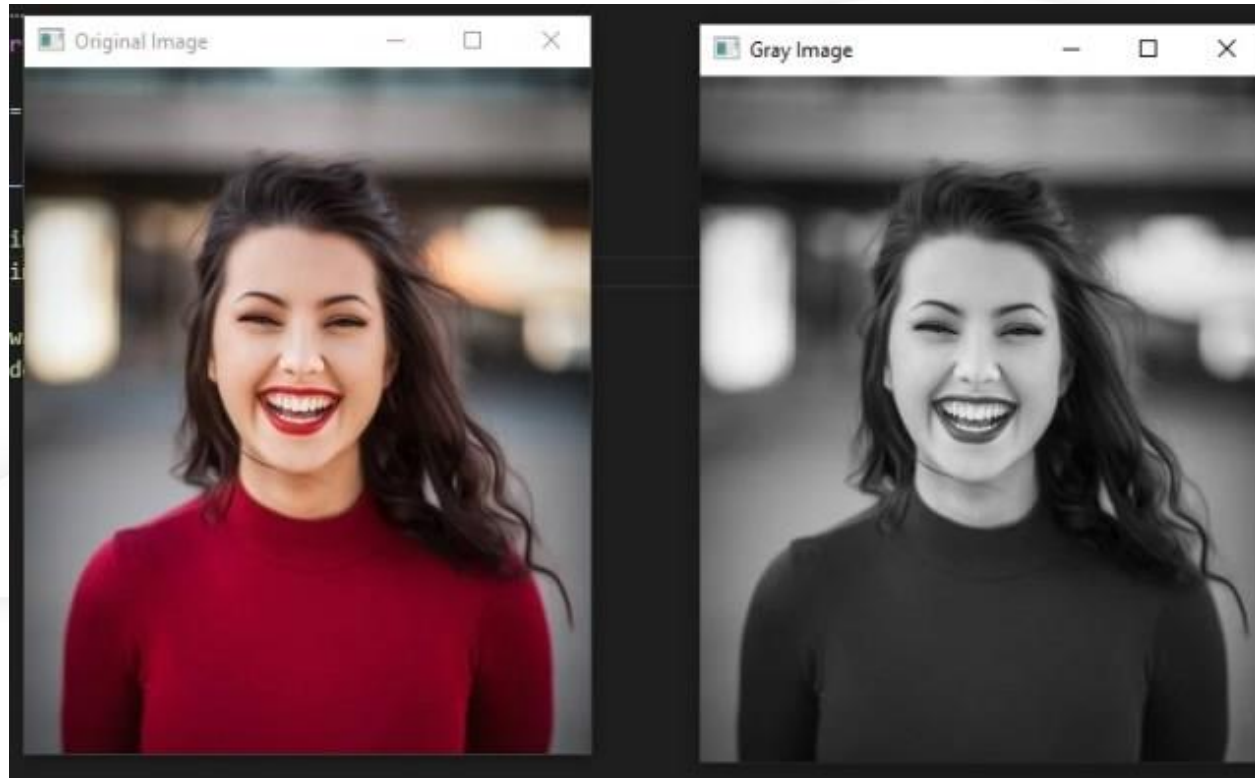
- Rescaling Image
- Correcting Illumination.

Domain	Input	Output
Image processing	Image	Image

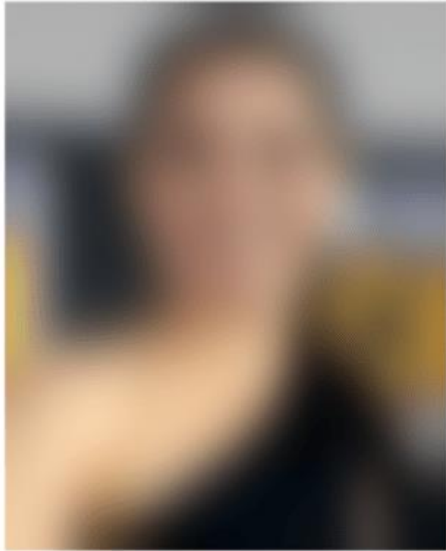
What is the effect of image processing ?



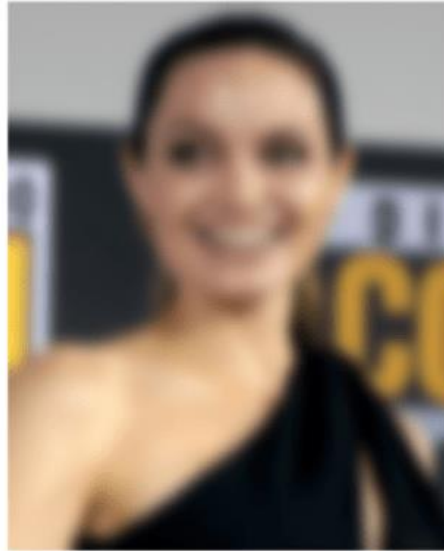
What is the effect of image processing ?



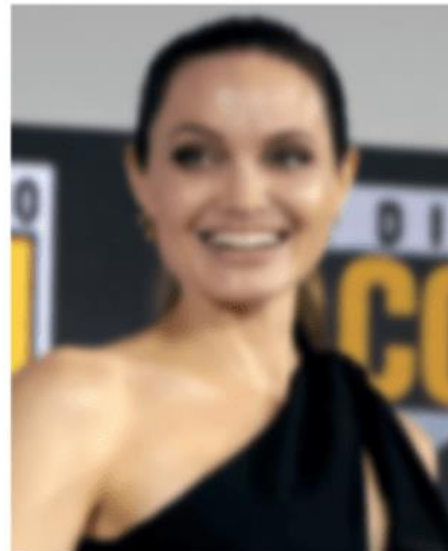
What is the effect of image processing ?



Factor = 3



Factor = 10



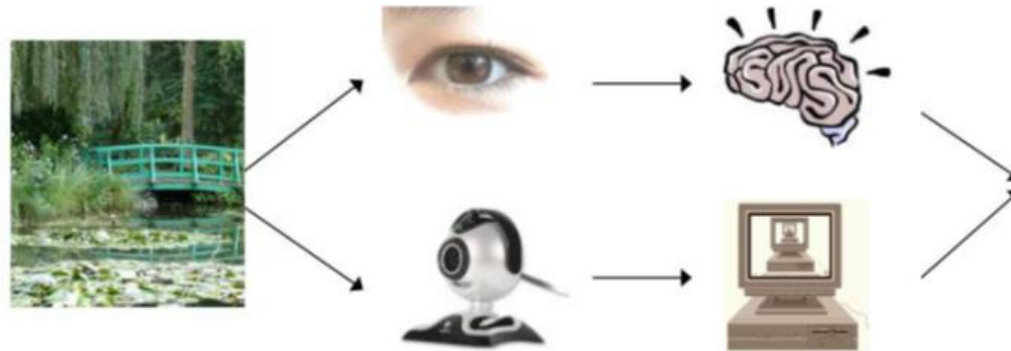
Factor = 20

What is the computer vision?

is a subject that enables the computer to understand the image and video collected by the acquisition device. Specifically, the computer is equipped with eyes (camera) and brain (algorithm) so that the computer can sense the environment.

Example of operations :

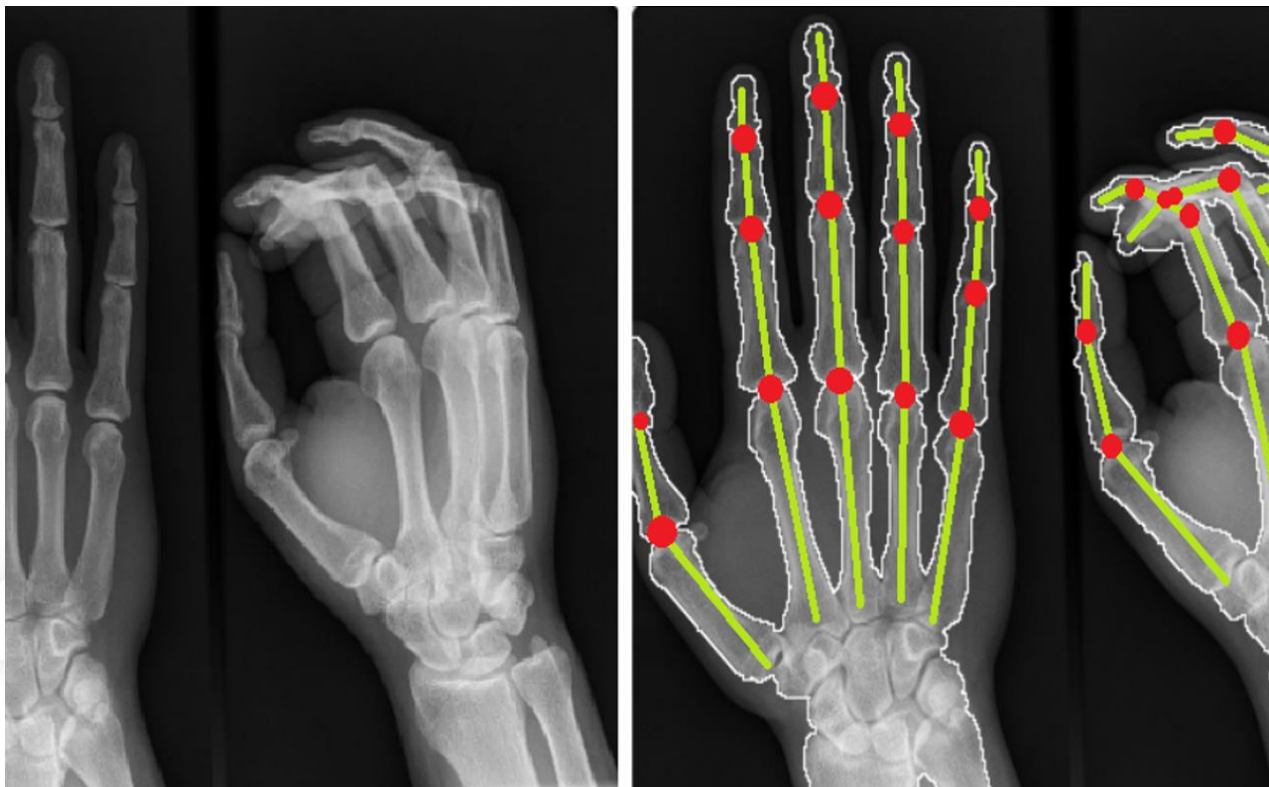
- Self-driving cars.
- X-Ray analysis.
- Cancer detection.



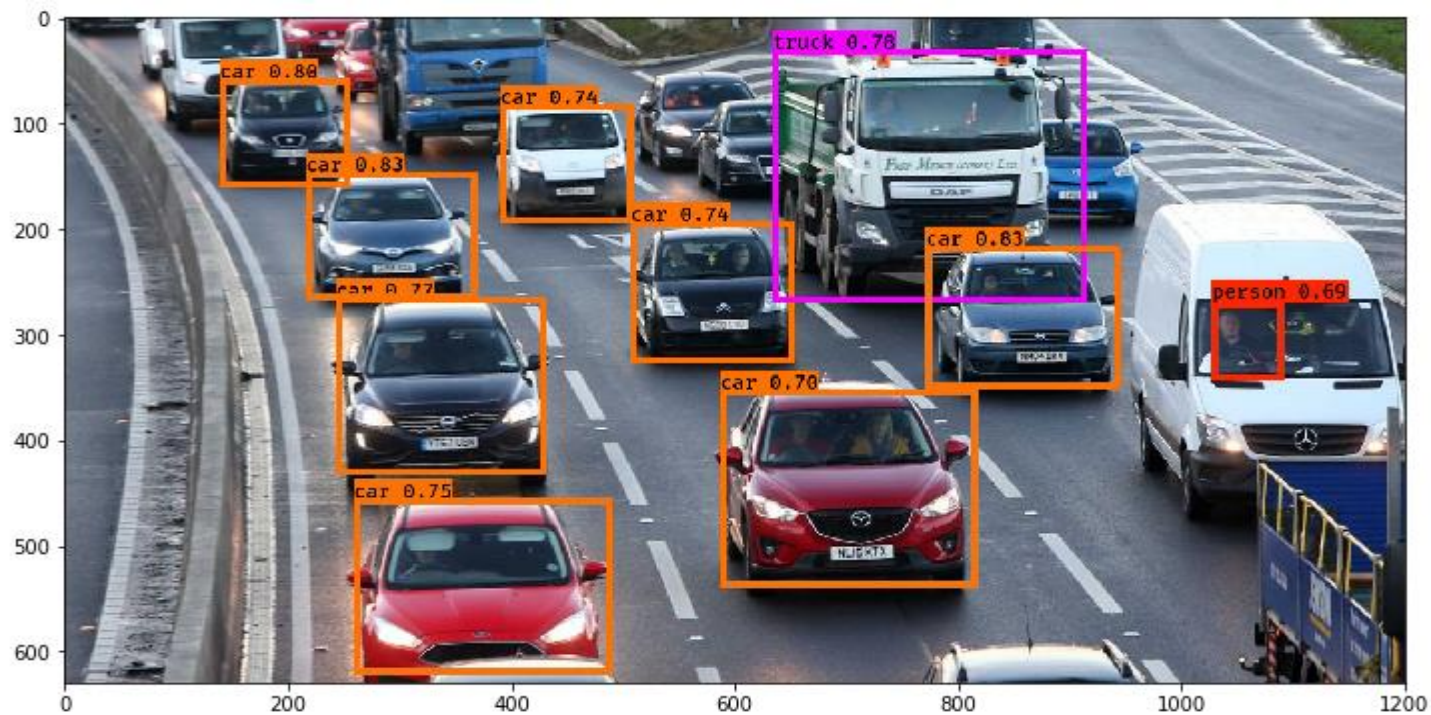
Self-driving cars



X-Ray analysis

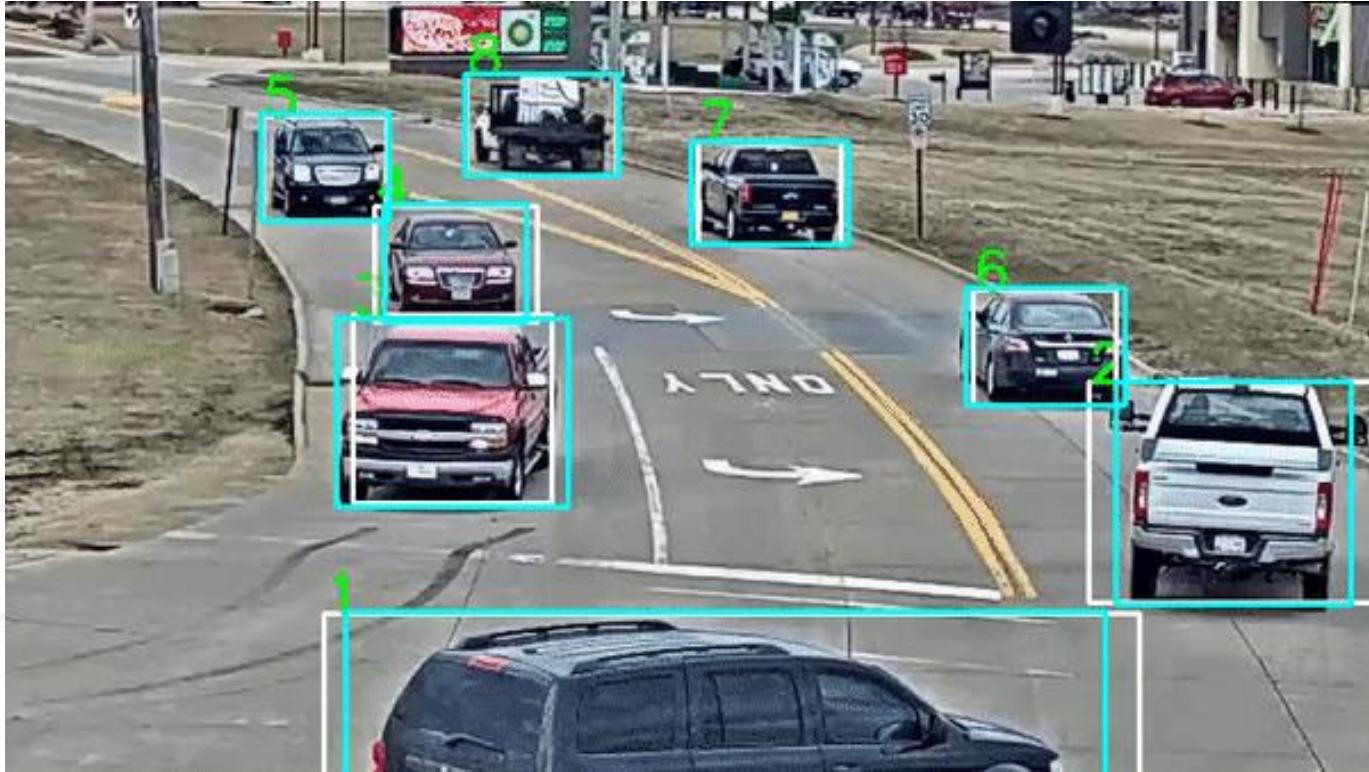


Camera Detection



How to benefit from computer vision in Traffic ?

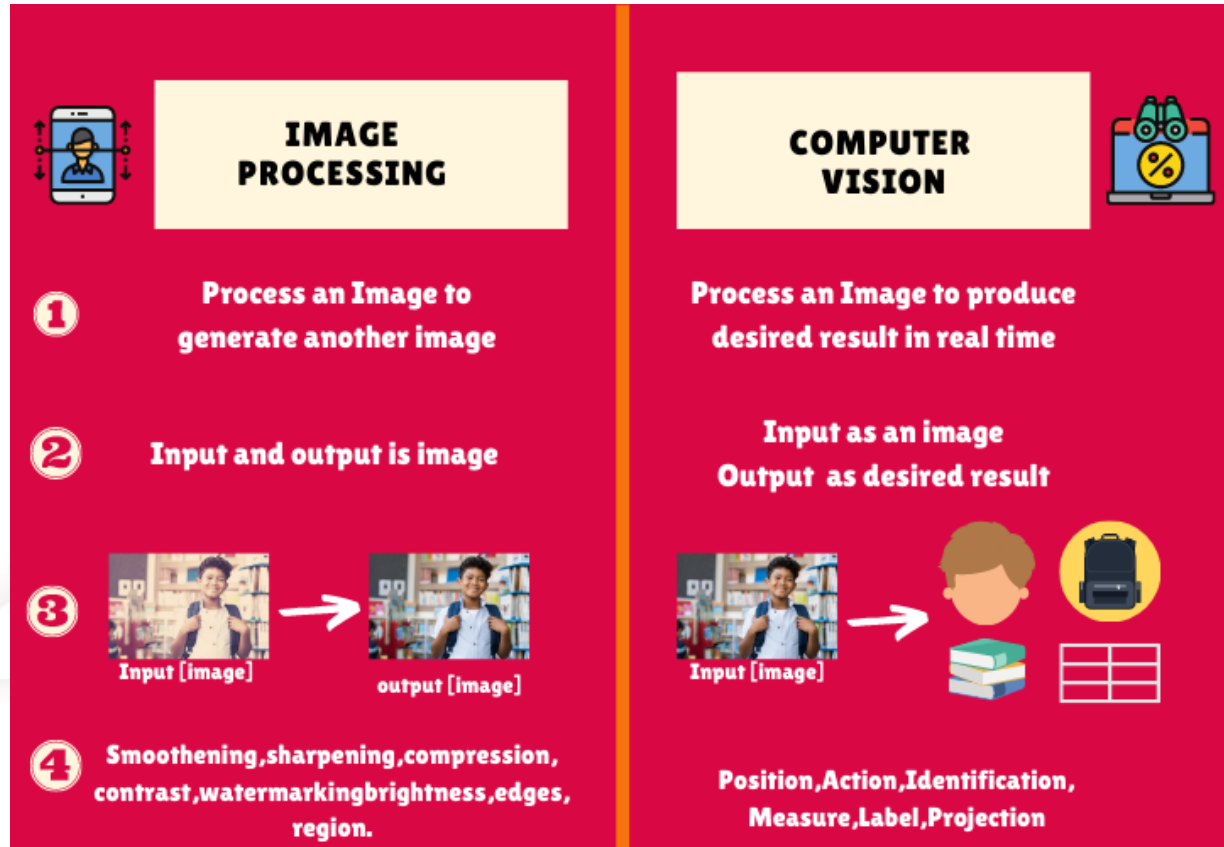




Could you tell me the difference between Image processing and Computer Vision?



What is the difference between Image Processing and Computer Vision?

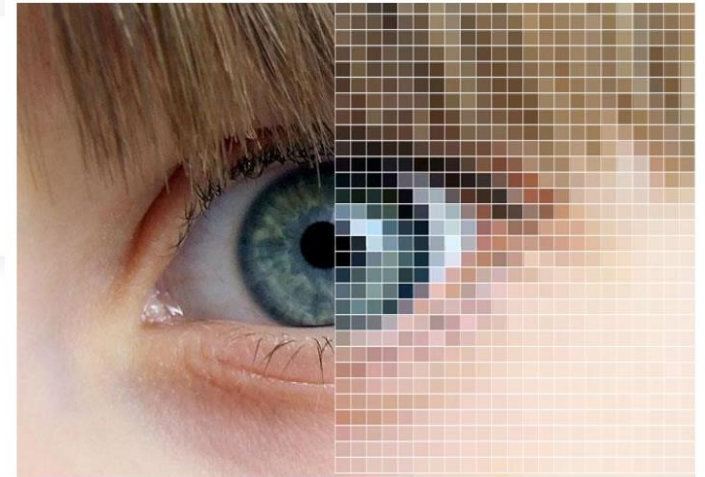


What is the definition of image ?

- An Image is formed as the result of projection of 3D scene into 2D .
- A rectangular grid of pixels and it has definite width and height counted in pixel.
- Each pixel (picture element) demonstrates the “intensity” of light or “color”.

What is the types of images ?

- Binary image.
- Grey image.
- Color image(RGB).



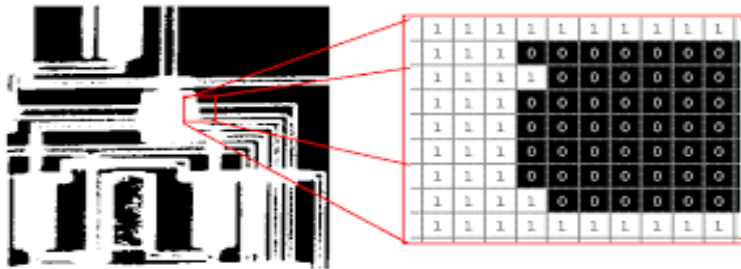
(1) Binary Image

A binary image is a black and white image where each pixel value is either 0 or 1. The value 0 represents background or black and the value 1 represents foreground or white.

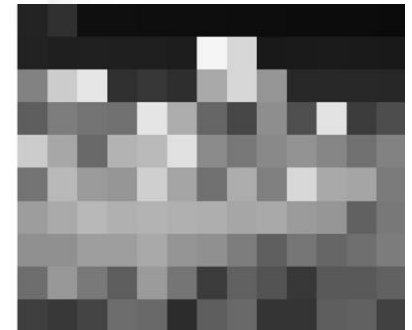
(2) Grayscale Image

A grayscale image is a black and white image with various shades of gray. The pixel value of a grayscale image is represented as an 8-bit integer. i.e., values between 0 and 255.

The pixel value of the grayscale image represents the brightness of the pixel. The value 0 is black, the value 255 is white, and the values in between makeup different shades of gray.



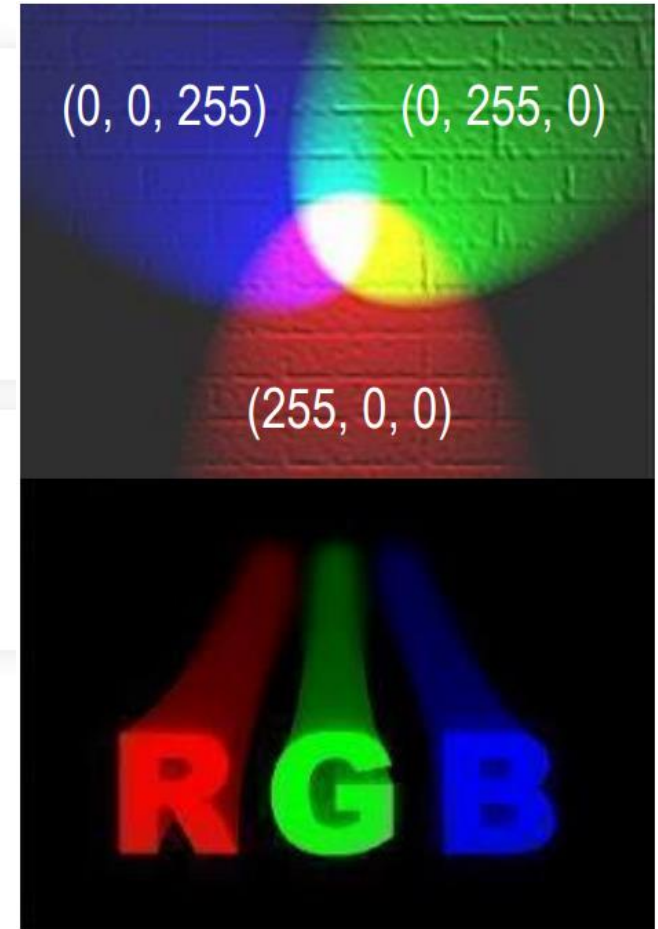
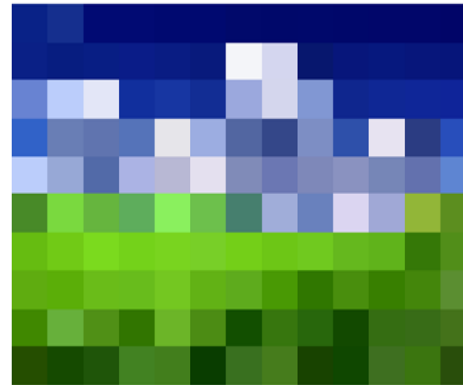
0	1	2	3	4	5	6	7	8	9	10	11	12
35	48	15	15	14	14	14	13	13	12	12	12	11
34	31	32	31	31	29	245	215	25	26	27	26	25
130	204	230	48	55	46	169	215	150	40	41	41	39
94	125	116	113	229	173	102	71	142	79	228	61	77
204	167	105	180	186	225	139	120	137	147	134	114	130
116	185	155	150	207	165	113	173	128	216	169	165	123
157	170	184	176	179	176	174	166	169	155	150	97	120
144	144	158	158	169	149	144	125	95	118	102	111	124
110	152	121	94	155	117	60	97	83	56	90	90	97
63	57	68	109	106	45	94	105	53	53	94	97	65



(3) Color image:

- is a mixed of three main colors: Red, Green, and Blue.
- used on a digital display screen.

	0	1	2	3	4	5	6	7	8	9	10	11	12
	125	142	115	115	112	112	107	106	107	106	106	106	104
	0	1	2	3	4	5	6	7	8	9	10	11	12
	24	47	10	10	0	0	0	0	0	0	0	0	0
0	1	2	3	4	5	6	7	8	9	10	11	12	
9	21	1	1	0	0	1	0	0	0	1	1	1	
10	7	8	10	9	8	244	212	7	7	6	7	6	
104	187	227	17	23	18	155	212	129	15	15	15	15	
49	105	96	85	230	156	82	52	125	46	231	43	40	
187	151	82	171	184	228	128	107	126	138	118	99	94	
72	122	102	94	138	109	70	160	105	219	160	146	92	
102	113	123	116	121	120	116	109	112	101	96	53	81	
95	91	106	104	116	98	94	73	48	73	56	68	91	
64	103	80	49	108	76	18	55	40	18	53	56	68	
37	21	31	66	66	9	57	70	24	15	62	59	41	



Types of images :

	Binary image	Grey image	Colored image
Numbers of bits	1 bit	8 bits	24 bits
Ranges	0 or 1	0 to 255	0 to 255 for each channel

color-spaces models:

The color spaces in image processing aim to facilitate the specifications of colors in some standard way.

Types of color-spaces models:

- RGB model
- HSV
- CMYK

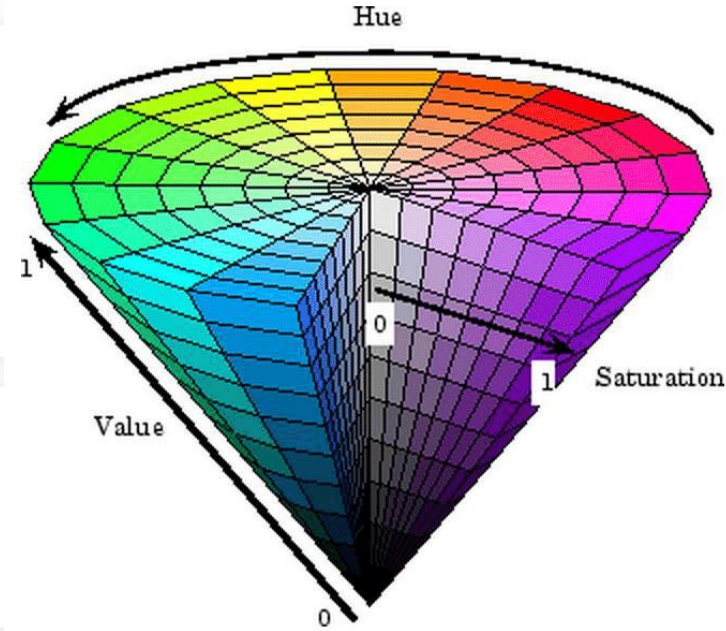
RGB:

is the most common color model used in [Digital image processing](#) and OpenCV. The color image consists of 3 channels. One channel each for one color. Red, Green and Blue are the main color components of this model.

HSV:

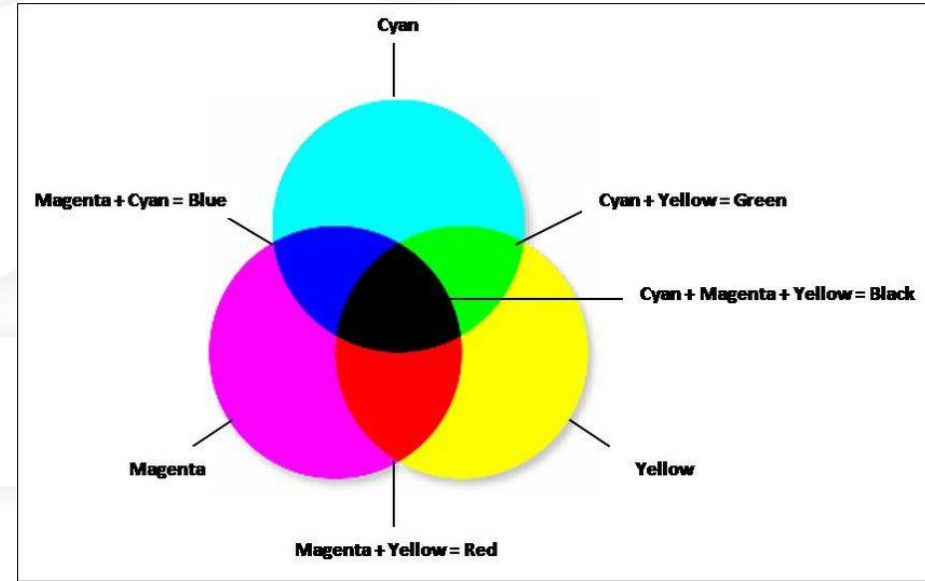
The image consists of three channels. Hue, Saturation and Value are three channels. This color model does not use primary colors directly. It uses color in the way humans perceive them. Used in [graphics such as photoshop](#).

- **H(hue)** is measured in degrees. ([color Type](#))
- **S(saturation)** indicates the ratio of the [purity](#) of the selected color to the maximum purity of the color.
- **V(value)** indicates the [brightness](#) of the color.



CMYK:

- color model is widely **used in printers**. It stands for Cyan, Magenta, Yellow and Black (key). It is a subtractive color model.
- $1 - \text{RGB} = \text{CMY}$
- Cyan is negative of Red.
- Magenta is negative of Green.
- Yellow is negative of Blue.



color-spaces models

	RGB	HSV	CMYK
COMPONENT	Red, Green, Blue	Hue, Saturation, Value	Cyan, magenta, yellow, key
USED IN	In Digital processing	In graphics such as photoshop.	In printers
NAME	an additive color model	an additive color model	subtractive color model
Collect all	White Color	proportional	Black Color

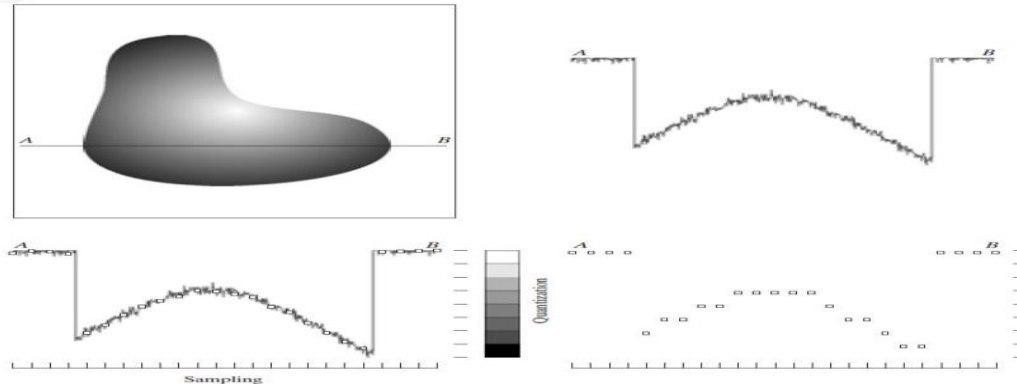
Image Digitization

The output of most sensors is a continuous voltage waveform, and image digitization is to **convert continuous images into discrete points** that can be processed by computers.

Image digitization involves two processes: **sampling** and **quantization**.

Image sampling: is to convert a spatially continuous image into discrete points.

Quantization: is the process of converting the **pixel grayscale** into discrete integer values



Coordinate Transformation:

The process of converting the coordinates in a map or image from one coordinate system to another.

Examples:

- Translation.
- Rotation.
- Scaling.
- mirroring.

Translation:

The point whose coordinates are $p = (x, y)$ in an image is moved to a new position by a translation vector (T_x, T_y) . The coordinates after transformation are $p' = (x', y')$. The transformation matrix of

translation is $A = \begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix}$.

Example:



Rotation:

Image rotation can be considered as rotation of pixels around an axis perpendicular to the image screen.

The point whose coordinates are $p = (x, y)$ in an image is rotated θ° clockwise around the axis that is located at the origin of coordinates. The coordinates after transformation are $p' = (x', y')$.

The rotation matrix is $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

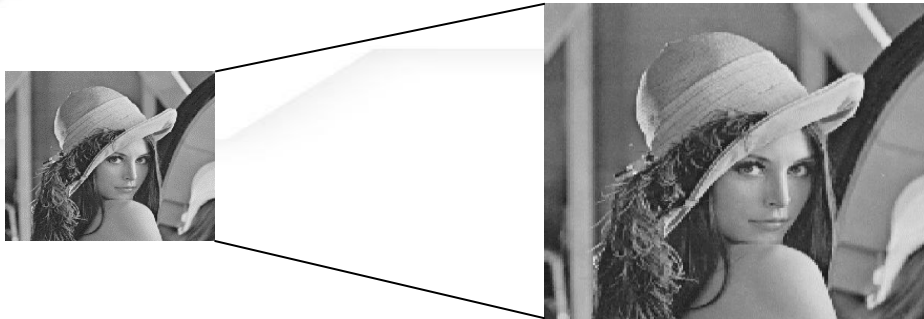


Scaling:

The resolution of the original image is (w, h) , and the image size is (w', h') after zooming. The zooming coefficient is $Sx = w'/w$ in the horizontal direction and $Sy = h'/h$ in the vertical direction.

The transformation matrix of zooming is $A = \begin{bmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Example:



Mirroring

The point whose coordinates are $p = (x, y)$ in a $w \times h$ image is mirrored with respect to the central axis of the image. The coordinates after transformation are $p' = (x', y')$.

The transformation matrix of horizontal mirroring is $A = \begin{bmatrix} -1 & 0 & w - 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

The transformation matrix of vertical mirroring is $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & h - 1 \\ 0 & 0 & 1 \end{bmatrix}$.



Original image



Horizontal mirroring



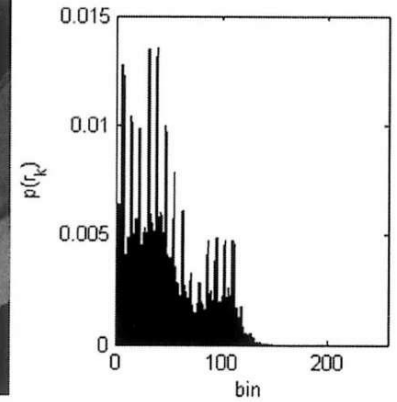
Vertical mirroring

Histogram:

- Capture the distribution of gray levels in the image.
- How frequently each gray level occurs in the image.



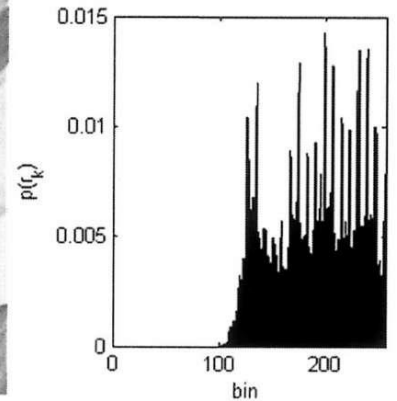
(a)



(b)



(c)

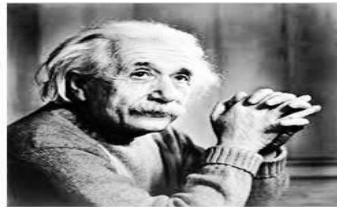


(d)

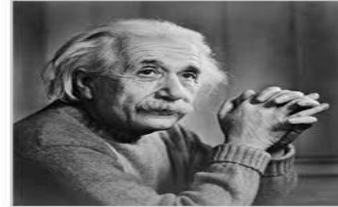
Histogram Equalization:

is a computer image processing technique used to improve contrast in images . It accomplishes this by effectively spreading out the most frequent intensity values, i.e., stretching out the intensity range of the image.

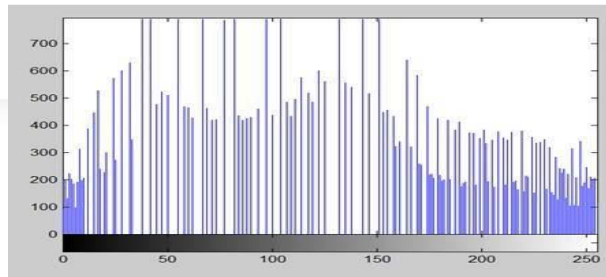
New Image



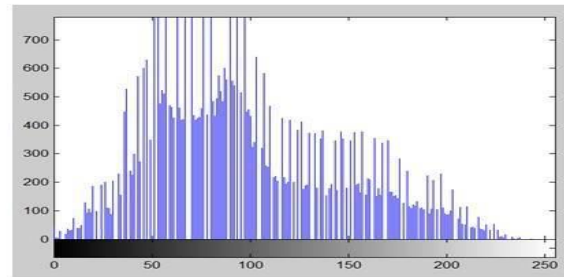
Old image



New Histogram



Old Histogram



Noise:

Noise in an image is the presence of artifacts that do not originate from the original scene content.

How to Remove Noise :

Filtering image data is a standard process used in almost every image processing system. Filters are used for this purpose. They remove noise from images by preserving the details of the same.



Mean Filtering :

The mean filter refers to the filter in which all template weights are 1. The average value of the neighborhoods of the pixel is used as the output result. Mean filtering can be used to [smooth the image and eliminate the noise](#). However, as the size of the template [increases](#), the image becomes [more blurred](#). It is often used to blur images.

$$\frac{1}{9} \times$$

1	1	1
1	1	1
1	1	1



Original image



3 x 3 mean filtering



5 x 5 mean filtering

Gaussian Filtering:

The Gaussian filter can be used to reduce the image blurring caused by increased template sizes.

1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1



Original
image



5 x 5 mean
filtering



5 x 5 Gaussian
filtering

Median Filtering

The median filter replaces the original pixel value with the median output of the pixel values sorted in the neighborhood. Median filtering implements [noise reduction and retains the sharpness of the original image](#), without changing the intensity value of the original image.

Median filtering is commonly used and can effectively suppress [salt-and-pepper noise](#) and protect the edge from blurring while suppressing random noise.



Original image



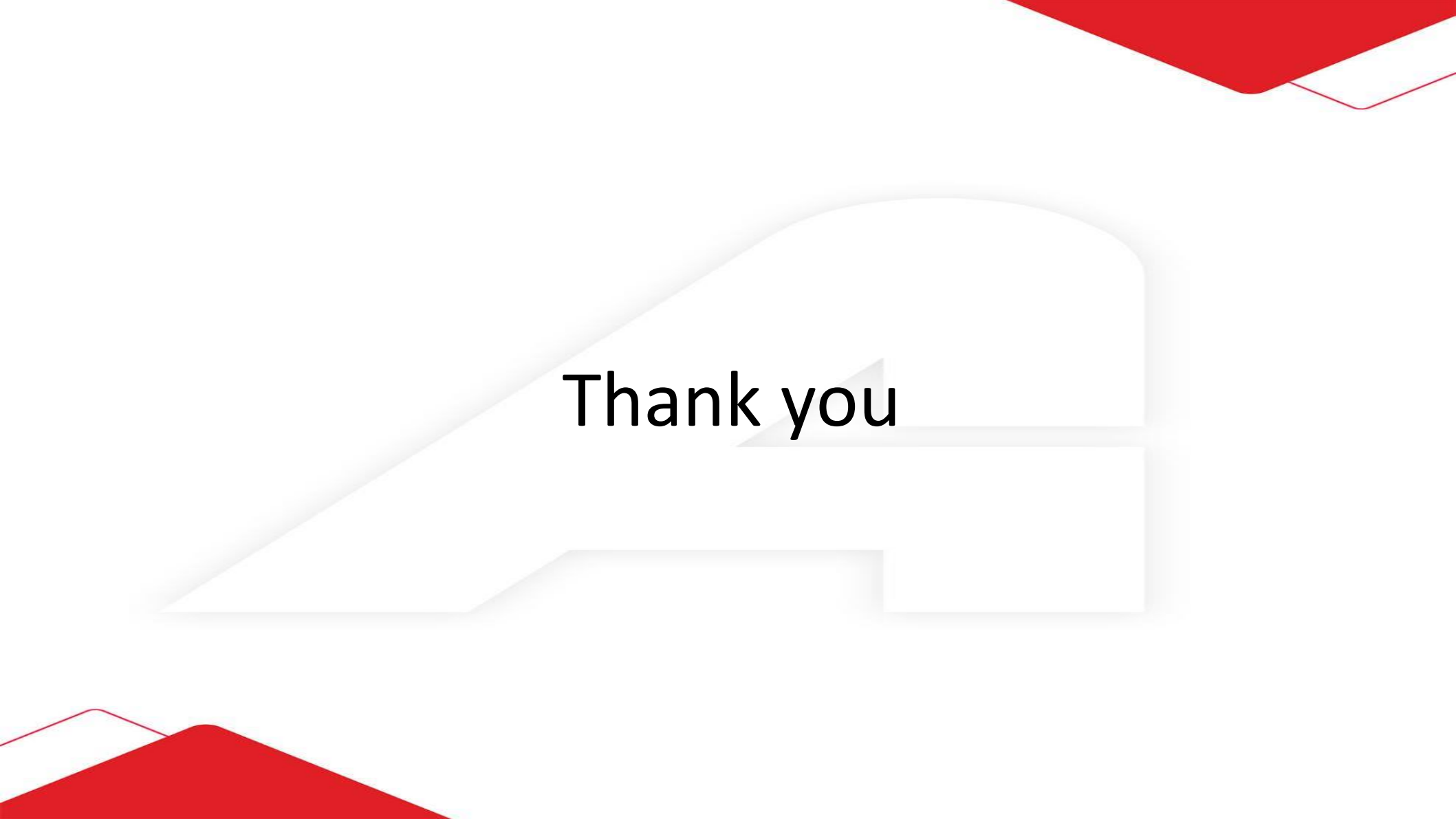
3 x 3 mean filtering



3 x 3 median filtering

Assignment

- 1- Read any image but using OpenCV Not scikit learning and apply coordinates Transformation(Translation- Rotation - Scaling).
- 2- search about perspective Transformation and apply it using OpenCv



Thank you